MILITARY SPECIFICATION

LINEAR SHAPED CHARGE ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for the design, development and preproduction testing of Linear Shaped Charge (LSC) Assemblies to be utilized for Aircrew Escape Systems. A shaped-Charge Assembly consists of a length of linear shaped charge supported by a housing (retainer) two booster tips(one at each end), and such fittings as are needed for attaching to other explosive assemblies.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-5541	Chemical Film and Chemical Film Material for Aluminum and Aluminum Alloys
MIL-I-6866	Inspection, Penetrant, Method of
MIL-I-6868	Inspection Process, Magnetic Particle
MIL-F-7179	Finishes and Coatings: General Specification for Protection of Aerospace Weapons Systems, Structures and Parts
MIL-S-7742	Screw Threads, Standard Optimum Selected Series, General Specification for
MIL-I-8500	Interchangeability and Replaceability of Component Parts for Aircraft and Missiles
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-I-8950	Inspection, Ultrasonic, Wrough Metals, Process For
MIL-T-18303	Test Procedures; Preproduction Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-A-22771	Aluminum Alloy Forgings, Heat Treated

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STANDARDS

Military

MIL-STD-130	Identification Marking of Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-453	Inspection, Radiographic
MIL-STD-794	Parts and Equipment, Procedure for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-889	Dissimilar Metals
MIL-STD-1166	Radiographic Testing Requirements for Solid Propellant
MIL-STD-1316	Fuzes, Navy, Design Criteria for
MS17983	Compass, Magnetic, Pilot's Standby

(Copies of specifications and standards required by suppliers in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents forms a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

STANDARDS

Navy

WS 4660	DIPAM Explosive (U) Purchase Description
WS 5003	HNS Explosive (U) Purchase Description

Industry

B46.1	Surface Texture

(Application for copies should be addressed to the American National Standards Institute (ANSI), 1430 Broadway, New York, N. Y. 10018).

3. REQUIREMENT

3.1 Preproduction. This specification makes provision for preproduction testing.

3.2 Section of Specification and Standards. Specifications and Standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.3 Material . All materials used in propellant formulation, hardware components, adhesives, etc. shall be in accordance with existing Government specifications. Use of materials not covered by existing Government specifications shall require prior approval by the cognizant design agency.

3.3.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889 shall not be used in contact with each other. Where such contact is unavoidable, the parts shall be protected in accordance with MIL-F-7179.

3.3.2 Aluminum. The following requirements shall apply.

a. Where a high strength aluminum alloy is required, 7075-T73 in accordance with specification MIL-A-22771, shall be used. (Not for fittings/ connector)

b. Ultrasonic inspection requirements for aluminum alloy products shall be in accordance with MIL-I-8950.

3.3.3 Magnesium and Magnesium Alloys, Magnesium alloys shall not be used.

3.3.4 Castings. The use of castings for structural applications require procuring activity approval prior to use.

3.3.5 Fungus proof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent.

3.4 Design & Construction. The LSC assembly shall be of the simplest and most efficient design consistent with the proposed use. Specific consideration shall be given to the factors of minimum size and weight while adhering to functional requirements, reliability and safety of operation.

3.5 Parts.

3.5.1 Residual Magnetism. All parts made of materials that are capable of retaining residual magnetism shall be demagnetized sufficiently to prevent component malfunction due to accumulation of magnetic contaminants.

3.5.2 Inspection of Metal Parts. All highly stressed steel parts shall be subjected to magnetic particle inspection in accordance with specification MIL-I-6868 or a similar Method. Titanium or aluminum high stress parts shall be inspected in accordance with MIL-I-6866, penetrant inspection.

3.6 Protective Treatment. All materials used which are not inherently corrosion resistant shall be treated in accordance with MIL-F-7179 to resist any corrosive effects resulting from the environmental conditions, specified herein. Any protective coatings used shall not crack, chip, scale or degrade due to the conditions specified herein.

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3.6.1 Aluminum. All aluminum and aluminum alloys shall be anodized in accordance with MIL-A-8625 Type II coatings except that in the absence of abrasive conditions, the anodize may be Type I coating where applicable, or may be chemical film in accordance with MIL-C-5541. (Not for the LSC assembly).

3.7 Surface Roughness. Surface roughness shall be as specified on the contractors drawings as outlined in B46.1.

3.8 Performance Reliability. The LSC assembly must meet all performance requirements at a reliability of not less than 99.9 percent at a confidence level of 90 percent over the temperature range of -65° to 350°F.

3.9 Weight (Guaranteed). The guaranteed weight of the equipment shall be kept to a minimum consistent with the requirements of this specification.

3.10 Self-Containment. The assembly shall be a self-contained unit, capable of inter-connection with similar units by means of shielded mild detonating cord (SMDC), explosive transfer lines, linear shaped charge assemblies or connect some functional device directly to the end of a shaped-charge assembly.

3.11 Sealing. The assembly shall be sealed to prevent moisture penetration into the area of the Linear Shaped Charge (LSC) apex prior to functioning.

3.12 Structural Integrity. When fired under any combination of the environments specified herein, the assembly shall exhibit sufficient structural integrity to prevent backblast or shrapnel from being generated in a manner that might create a personnel hazard within the cockpit area with all internal explosive charges increased 25% over the maximum loading tolerance.

3.13 Threads. All threads shall be in accordance with Specification MIL-S-7742.

3.14 Junction Fittings. The assemblies shall incorporate junction fittings having threaded ports to receive the initiating SMDC or LSC. These threaded ports shall be capable of withstanding installation of 90 in-lb. torque in accordance with the following schedule:

3/8-24-UNF-3A THD.
7/16-20-UNF-3A THD.
1/2-20-UNF-3A THD.
5/8-18-UNF-3A THD.

Those fittings shall remain intact in port areas after firing of the shaped charge.

3.15 Lockwiring. Wherever LSC fittings have lockwire provisions the junction fittings should have lockwire provisions. Self-locking nuts are preferable for LSC assemblies.

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3.16 Explosive.

3.16.1 Linear Shaped Charge (LSC). The assembly shall incorporate Linear Shaped Charge (LSC) as the explosive cutting medium. Composition of the LSC shall include an explosive core of hexanitrostilbene (HNS-II), or chemical compound dipicramide (DIPAM Grade A) in accordance with WS4660 Grade A, contained within an aluminum sheath. The HNS shall fully satisfy the requirements of WS5003 and MIL-STD-1316.

3.16.2 Standoff. The LSC shall be contained within a flexible housing strip whose dimensions are such that, when the face of the housing strip is in contact with the surface to be cut, the LSC is positioned at its optimum standoff distance.

3.16.3 Output. The explosive output of the LSC shall be sufficient to meet the detailed requirements.

3.17 Explosive transfer. The contractor shall submit details of interface alignment for all discontinuous points of the LSC assembly. Detonation transfer will be subjected to analysis and approval by the procuring agency.

3.18 Inspection - Methods.

3.18.1 General. Any voids, cracks, discontinuities, construction defects, misalignment, or the absence of a component part discovered by any of the inspection techniques described in this section shall be cause for rejection.

3.18.2 Radiographic.

3.18.2.1 X-Ray Inspection. All assemblies containing explosive interfaces shall be radiographically inspected in accordance with Standard MIL-STD-453 in order to confirm the presence, alignment, density, proper positioning and integrity of all explosive components.

3.18.2.2 Neutron Radiographic Inspection. The assemblies shall be neutron ray inspected along their full length prior to forming. The accept/reject criteria to be used in the evaluation of the radiographs shall be coordinated between the contractor and the procuring activity and shall be approved prior to qualification. Radiographic film shall be retained for a minimum of 7 years.

3.18.3 Surface. Each component shall be inspected for surface and sub-surface discontinuities. Where appropriate a magnetic particle inspection shall be performed in accordance with MIL-I-6868B or a similar method. For non-ferrous metals, a penetrant method of inspection shall be performed in accordance with MIL-I-6866.

3.18.4 Leakage Requirements. The assemblies shall be sealed to prevent penetration by water, moisture, and dust with a maximum leakage rate of 1×10^{-5} std cc/sec of air at a pressure differential of 1 ± 0.1 atmosphere. All gases

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produced by actuation shall be contained to preclude deleterious effects on aircraft systems and occupants.

3.18.5 LSC Control. A control sample of LSC shall be taken at the beginning and end of each production run with at least one sample being taken every 100 feet of manufactured length. Tests shall be performed on all such control samples in order to verify the following performance capabilities:

a. The LSC shall meet the requirements of MIL-STD-1316.

b. The minimum detonation velocity of the LSC shall be 6000 meters per second.

c. The maximum moisture content of the explosive compound when loaded into the LSC shall be 0.2 percent of weight.

d. The core loading tolerance shall not exceed ± 10 percent of the nominal core load.

3.19 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable as defined in MIL-I-8500.

3.20 Identification and Markings. Identification and markings shall be in accordance with MIL-STD-130.

3.21 Workmanship. Workmanship shall be in accordance with high grade aircraft practice and of quality to assure safety, proper operation and service life. Workmanship shall be subject to inspection and approval by the sellers quality control.

3.22 Environmental Conditions.

3.22.1 Standard Conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions for making laboratory bench tests:

Temperature	+70°F ± 5°F
Altitude	Normal Ground
Vibration	None
Humidity	90% maximum

3.22.2 Service Conditions. The assembly shall operate under the following environmental service conditions or combination of these conditions as specified herein.

3.22.3 Temperature. The assembly shall be capable of satisfactory operation during and after prolonged exposure to the following temperatures.

(a)	Storage	-80° to +160°F
(b)	Service	-65°F to +350°F
(c)	Peak	400°F for periods of five minutes

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3.22.4 Drop.

3.22.4.1 Forty-Foot Drop. The assemblies shall not initiate and shall be safe to handle and dispose of when dropped from a height of 40 feet. The assemblies are not required to operate after being exposed to this condition.

3.22.4.2 Six-Foot Drop. The assemblies shall not detonate and shall be capable of satisfactory operation after being subjected to the test specified in 4.5.1.3.

3.22.5 High Temperature. The assemblies shall be capable of withstanding the temperatures as specified in 4.5.1.2.8 and shall meet the design performance requirements when test fired.

3.22.6 Auto Detonation. This is a safety test designed to assure that the system will not be initiated by thermal exposure. This test shall be conducted as specified in para. 4.5.1.2.6.

3.22.7 High temperature exposure (12 hours). The maximum temperature (within 25°F) shall be established to which LSC may be exposed for 12 hours and function within design performance requirements when fired after cooling to 70°F. The maximum temperature shall be determined by the high temperature exposure test as specified in 4.5.1.2.7.

3.22.8 Temperature-shock/humidity/altitute. The assemblies shall be capable of withstanding temperature shock/humidity/altitude cycling as specified in 4.5.1.2.1 and shall subsequently meet the design performance requirements when test fired.

3.22.9 Vibration. The assemblies shall not fire when subjected to vibration as specified in 4.5.1.2.2. After subjection to such vibration, the assemblies shall meet the design performance requirements when test fired.

3.22.10 Shock. The assemblies shall be capable of withstanding shock conditions with peak amplitudes of 20 g for a duration of 11 ms as specified in 4.5.1.2.5 and shall subsequently meet the design performance requirements when test fired.

3.22.11 Dust. The assemblies shall be capable of withstanding exposure to dust-laden atmosphere as specified in 4.5.1.2.4 and shall subsequently meet the design performance requirements when test fired.

3.22.12 Submerged firing. The assembly shall shall be capable of firing while immersed in water and at a pressure of 60 psig.

3.23 Product marking. Each assembly shall be clearly and permanently identified with the LSC designation, identifying number, lot number, loading date (month and year), and manufacturer's identification symbol.

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3.24 Foolproofness. The assemblies shall be designed to prevent incorrect assembly and installation.

3.25 Ruggedness. The LSC shall withstand ordinary handling and manipulation by unskilled personnel throughout the course of shipping, inspection, installation and service.

3.26 Performance Reliability. The assemblies must meet all performance requirements at a reliability of not less than 99.9 percent at a confidence level of 90% over the temperature range of -65°F to +350°F.

3.27 Inadvertent fire probability. The inadvertent fire probability of the LSC shall be zero under any environmental conditios specified herein. Any evidence that the LSC could fire prematurely shall be cause for rejection of the design. An inadvertent firing is defined as an involuntary, or premature ignition of any explosives in the LSC.

3.28 Life Requirements. The LSC shall have a total life eight years minimum including at least 5 years of service life. During this time, the LSC shall not suffer deterioration or loss of performance beyond the limits specified herein.

3.29 Storage Life. Storage life is defined as that period during which LSC are stored in their original factory-sealed storage container, and begins with the date of manufacture. Under these conditions the temperature will not exceed the limits of -80°F to +160°F. The LSC shall be capable of being stored in any attitude.

3.30 Service Life. Service life is defined as the period of time beginning when the LSC storage container is unsealed and opened, whether the LSC is installed or not. During this period the LSC may experience any combination of environmental conditions specified herein.

3.31 Performance.

3.31.1 General. Each assembly shall provide an explosive output sufficient to completely sever an 0.063 inch thickness of 2024-T81 aluminum.

3.31.2 Initiation. The assembly shall be capable of being initiated by the donor SMDC. Each assembly shall contain two separate points of initiation.

3.32 Development testing. Development tests are conducted to provide data to be used in the design or in support of the design of a specific component or subsystem. Development tests may be used to determine operating characteristics under off-design conditions. Stress to failure development tests shall be conducted to provide failure mode characteristics for verification of analysis and determination of strength and operating margins of safety. It is

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required that development tests be run on the LSC under simulated thermal environment to assure the design compatibility with the operating requirements of this specification. Development tests shall be categorized as follows:

(a) Design Feasibility Tests (3.32.1)
(b) Design Verification Tests (3.32.2)

3.32.1 Design Feasibility Tests (DFT). Design feasibility tests shall include all tests conducted for the following purposes:

(a) Component and part selection

(b) Investigation of the performance of the LSC shall include as a minimum, performance or functional tests, to indicate rupture or fragmentation, firing and velocity, shock, vibration, humidity, operating margins, minimum charge (pyrotechnic), nominal flight charge and structural integrity.

(c) Demonstration of safety and operating margins or other analytical assemptions.

(d) Design Feasibility Tests results will establish the criteria for acceptance/rejection limits of detrimental deformation.

3.32.2 Design Verification Tests (DVT). Design Verification Tests are advanced stage development tests which are performed on the LSC for the purpose of substantiating the design for its intended mission. These tests shall be conducted under simulated mission environments and off-design conditions and shall include, as a minimum: shock, vibration, humidity, firing and velocity tests, structural integrity tests and X-ray and N-ray inspection. Successful completion of design verification tests provide the assurance to permit a design freeze decision to be made. The test shall use typical LSC permanently identified and numbered for the purpose of design verification tests. The contractor shall have available for review by the procuring activity all detail, in-process and final inspection records for each design verification LSC.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, if approved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Materials. Inspection and testing of component parts and assemblies shall be made to determine compliance with 3.3. Where defects, incompatibility, or inferior quality is evident, and the Government deems material analysis necessary, the contractor will be requested to submit samples or specimens to the cognizant design agency for analysis and approval.

4.2 Classification of tests. The inspection and testing of LSC shall be classified as preproduction and special tests.

4.3 Test conditions.

4.3.1 Measurements and records. All functional tests of LSC shall be performed with the equipment mounted and loaded in a manner to simulate service conditions as closely as possible.

4.3.2 Instrumentation. Any state-of-the-art-type instrumentation and recording equipment may be used; e.g., oscillograph, oscilloscope, or magnetic tape. Transducers may be any state-of-the-art type; e.g., piezoelectric, strain gage, variable reluctance, capacitive, or potentiometer. However, accuracy of all test apparatus shall conform to the requirements of MIL-STD-810. Documented calibration records will be maintained and be available for inspection by the cognizant design agency.

4.4 Test Methods.

4.4.1 Component Inspection. Each part of the assembly shall be 100 percent inspected for conformance to the applicable drawing.

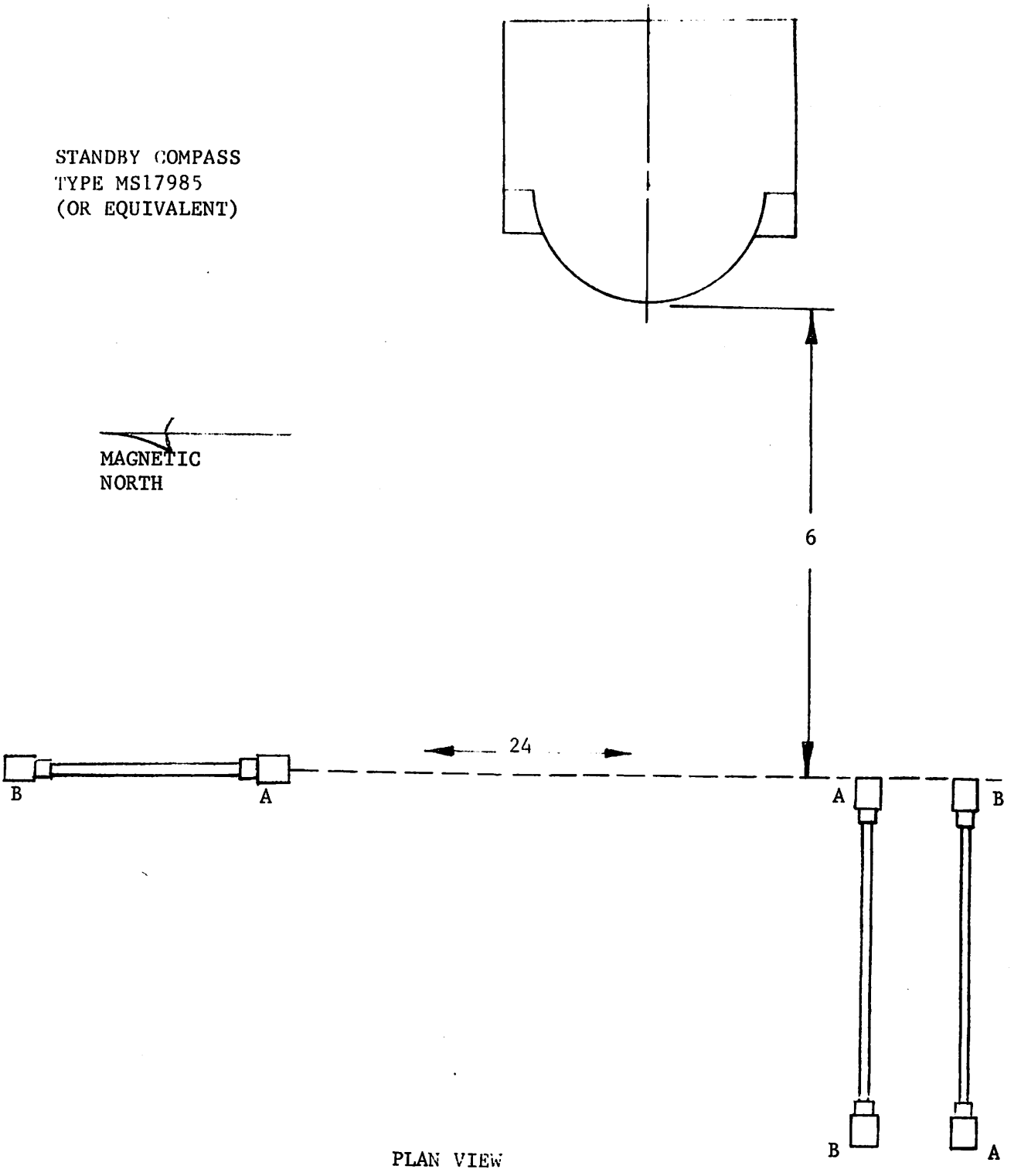
4.4.2 Examination of Product. Examination of the assemblies shall be made to detect visible defects and to determine that they meet the finish, and marking requirements, and to verify that they conform to the applicable drawing.

4.4.3 Leak Detection. All assemblies shall pass a dry gas leak test. Assemblies which exhibit a leak rate in excess of 10^{-5} cc per second of air at a pressure differential of 1 $\pm$ 0.1 atmosphere shall be considered defective.

4.4.4 Demagnetization Test. The assembly shall be checked for residual magnetism in an area free of local magnetic effects by placing the device 6 inches from a pilot's standby compass, type MS17983 or equivalent, in a north-south horizontal position with the compass placed in an east-west heading as shown in Figure 1. The device shall be placed at the same height from the floor as the compass in a longitudinal direction along its entire length. This procedure shall be repeated for every 90 degrees rotation of the device. The assembly shall not deflect the compass more than three degrees.

4.4.5 X-Ray Inspection. The assemblies shall be X-ray inspected along their straight length and ends only after forming, in accordance with MIL-STD-453. The accept/reject criteria to be used in the evaluation of the X-ray shall be coordinated between the contractor and the procuring activity prior to commencement of production.

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NOTE:

1. Assembly and compass shall lie in a horizontal plane.

FIGURE 1 Layout for Determining of Residual Magnetism in Assembly

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4.4.6 Neutron Radiographic Inspection. The assemblies shall be neutron ray inspected along their full length prior to forming in accordance with MIL-STD-1166. The accept/reject criteria to be used in the evaluation of the radiographs shall be coordinated between the contractor and the procuring activity.

4.5 Preproduction Tests. Preproduction tests shall be made on equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the approved test procedure of 4.7.

NOTE: When preproduction tests are to be made on equipments manufactured in the first production run, the equipments to be tested shall be selected by a representative of the procuring activity with the concurrence of the Government Inspector.

4.5.1 Scope of Preproduction Tests. Preproduction tests shall include all tests specified herein and shall be conducted in accordance with the requirements of this specification. Test values and parameters shall be as designated in the design requirements paragraphs of this specification. If during the course of conducting preproduction tests corrective action is required, the contractor shall re-evaluate the previously approved tests, test plans, and test procedures to assure their continued validity. The corrective action, extent of required retest, and the test plan and procedure revisions shall be subject to approval of the procuring activity. Preproduction tests shall be performed on 82 LSC assemblies in the following sequence and quantities shown in Table I:

Tests	Paragraph Numbers
Functional Test	4.5.1.1
Environmental Test	4.5.1.2
Forty-Foot Drop Test	4.5.1.3
Six-Foot Drop Test	4.5.1.4

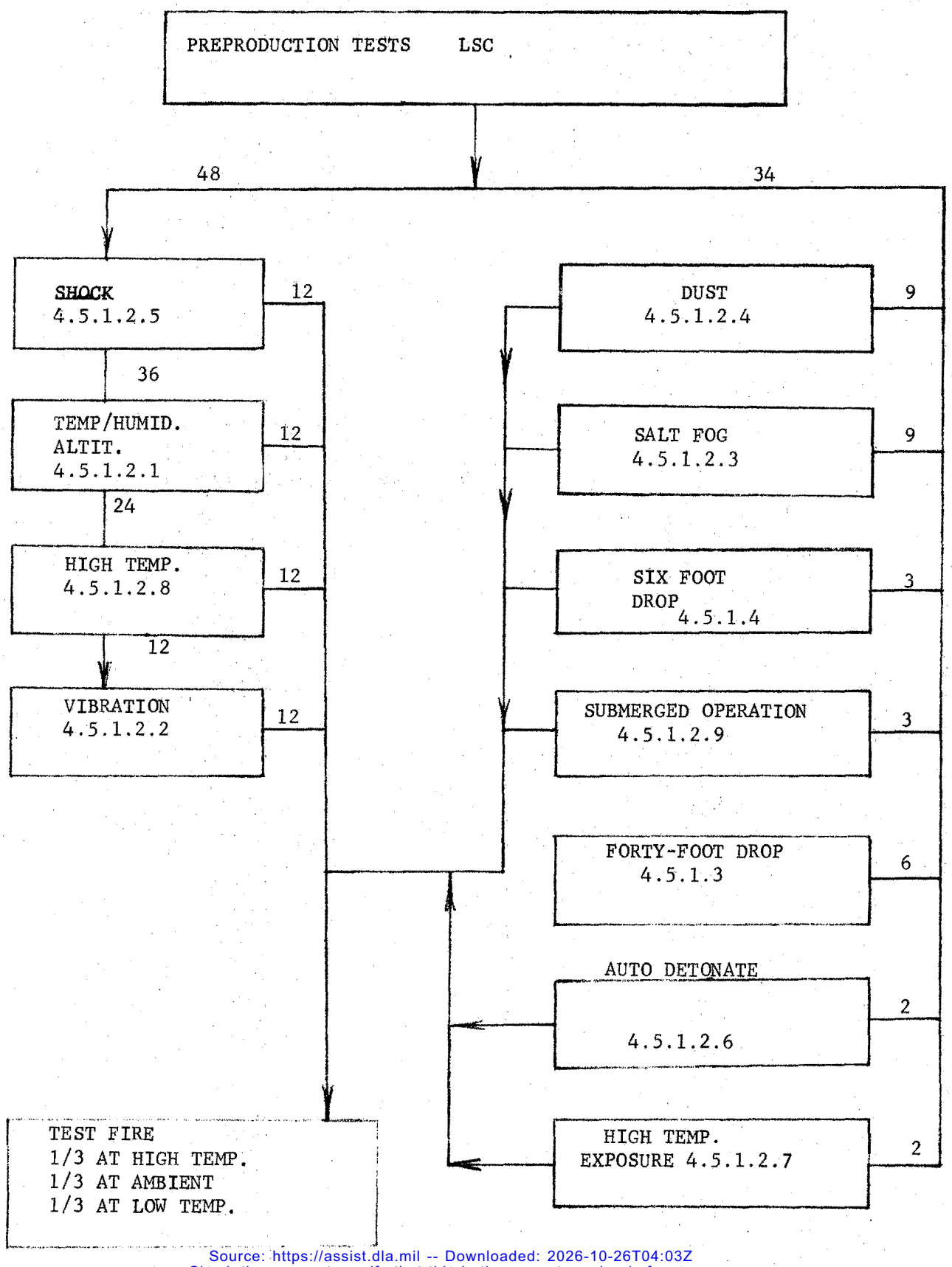
4.5.1.1 Functional Test

4.5.1.1.1 LSC Test. A sample length of LSC shall be tested for detonation velocity, moisture content, and core loading for each 100 feet of length manufactured. Acceptable values shall be as specified in 3.18.5. The LSC source, the sample length and the sample location shall be specified in the preproduction test report.

4.5.1.1.2 Temperature Conditioning for Firing. When specified in Table I the assemblies shall be placed into a thermal environmental chamber at atmospheric pressure and conditioned as follows:

(a). High Temperature. The temperature of the LSC under test shall be raised to +350°F and maintained at this temperature for a minimum of one hour after the assembly temperatures have been stabilized. The assemblies shall be fired while stabilized at this temperature.

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TABLE I - PREPRODUCTION TEST SEQUENCE

(b). Ambient Temperature. These tests shall be conducted at +70°F ± 5°F and fired while stabilized at this temperature.

(c). Low Temperature. The temperature of the LSC under test shall be cooled to -65°F for a minimum of one hour. When specified the assemblies under test shall be fired while stabilized at this temperature.

4.5.1.2 Environmental Tests. The following environmental tests shall be performed on the number of LSC's specified in Table I to perform the firing function. The contractor may start the next environmental sequence while conducting firing tests of the previously withdrawn samples. In no case shall the second subsequent environmental test be started without prior approval of the cognizant design agency.

4.5.1.2.1 Temperature Shock/Humidity/Altitude.

4.5.1.2.1.1 Temperature Shock/Humidity/Altitude Test. The assemblies shall be subjected to the temperature shock/humidity/altitude test as outlined below. The schedule has been arranged in such a manner that operations are not required outside normal working hours except for such supervision as may be necessary to insure proper operation of the test equipment. It is not mandatory that the day/clock time schedule given below be followed: however, it is mandatory that the time, environmental, and sequence requirements be adhered to. It is also noted that only two conditioning chambers are required to accomplish this test. No less than two chambers may be used and still accomplish the temperature shock portion of this test. It is permissible to use three chambers provided the time, environmental, and sequence requirements are met. The fluctuations from the specified temperatures shall not exceed 5°F. When the temperature/altitude chamber door is opened to place test items inside, the chamber pressure will become atmospheric. The time required to return the chamber pressure to 0.65 pounds per square inch (psi) shall not exceed 1 hour. Test specimens shall be supported on screen trays or racks so that all areas are exposed to the prescribed atmospheric conditions at all times throughout the test.

Procedure

Monday 0800 - Place test items in a chamber maintained at +70°F at 50 percent relative humidity (RH).

1200 - Raise chamber temperature to +160°F and the RH to 95 percent. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi)

Tuesday 0800 - Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.

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1200 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (o.65 psi).

1600 - Remove test items from above chamber and immediately place in a chamber maintained at +160°F at 95 percent RH.

Wednesday 0800 - Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900.

1200 - Raise chamber temperature to +160°F at 95 percent RH. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi).

Thursday 0800 - Remove test items from above chamber and immediately place in a chamber maintained at +70°F at 50 percent RH.

1200 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at a pressure altitude of 70,000 feet. (0.65 psi).

1600 - Remove test items from above chamber and immediately place in a chamber maintained at +160°F at 95 percent RH.

Friday 0800 - Reduce chamber temperature to +70°F at 50 percent RH. The chamber temperature shall reach +70°F at 50 percent RH not later than 0900.

1200 - Raise chamber temperature to +160°F at 95 percent RH. The chamber temperature shall reach +160°F at 95 percent RH not later than 1300.

1600 - Remove test items from above chamber and immediately place in a chamber maintained at -65°F at standard ambient pressure. This schedule shall be followed for a total of 4 weeks (28 days) except that on the second and fourth weekends the soak time shall be from 1200 on Friday until 0800 on Monday at a temperature of +160°F at 95 percent RH. At the conclusion of the temperature-shock/humidity/altitude test, the test items shall be allocated to the tests specified in Table I.

4.5.1.2.1.2 Test Firing. Upon completion of the temperature shock/humidity/altitude test, LSC shall be fired within four hours after removal from the environmental chamber.

4.5.1.2.2 Vibration Test. The test shall be conducted in accordance with Method 514 (aircraft category, procedure I, parts 1, 2 and 3 and curve H) of MIL-STD-810, except that for each resonant and cycling period the assemblies

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shall be divided equally for vibration at -65°F, and 70 and +350°F. After vibration testing has been completed the LSC assemblies shall be test fired.

4.5.1.2.3 Salt-Fog.

4.5.1.2.3.1 Salt-Fog Test. Prior to the salt-fog environment six LSC's shall have end caps installed, these six LSC's and the inert connectors required for that firing shall be subjected to all the environments and fired during the submerged operation test specified in 4.5.1.2.9. The test shall be conducted in accordance with Method 509 of MIL-STD-810.

4.5.1.2.3.2 Test Firing. Upon completion of the salt-fog test, LSC shall be fired within four hours after removal from salt-fog chamber.

4.5.1.2.4 Dust. This test shall be conducted in accordance with Method 510 of MIL-STD-810 except that the high temperature shall be 350°F. After completion of this test, the specimens shall be test fired.

4.5.1.2.5 Shock Test. This test shall be conducted in accordance with Method 516, procedure I of MIL-STD-810. The shock pulse waveform shall be a terminal peak saw tooth. The peak amplitude shall be 20g and the duration shall be 11 ms. The shock test specimens shall be divided into three groups and fired at -65°F, +70°F, +350°F.

4.5.1.2.6 Auto Detonation Test. Two LSC's shall be placed in an oven preheated to 400°F for one hour. If no LSC's auto detonates at high order during one hour, the test is complete. If high order detonation occurs testing shall be terminated and the reason(s) for the failure determined and corrected.

4.5.1.2.7 High Temperature Exposure (12 hours). Two LSC's shall be placed in an oven preheated to a temperature of +400°F maintained for 12 hours. If no LSC's auto detonate the two LSC's shall be cooled to 70°F and functionally tested. If any LSC's auto detonate or fails to meet design performance requirements after cooling, the test shall be repeated with additional groups of LSC's, decreasing the temperature in increments of 25°F, until design performance requirements are met.

4.5.1.2.8 High Temperature Test. This test shall be conducted in accordance with Method 501, procedure I of MIL-STD-810 except that the LSC's shall be exposed to a temperature of +350°F until thermal stabilization is reached and shall be maintained at that temperature for a period of 50 hours.

4.5.1.2.8.1 Test Firing. Upon completion of the high temperature test, assemblies shall be fired within 5 minutes after removal from high temperature chamber.

4.5.1.2.9 Submerged Operation Test. The 3 LSC assemblies shall be submerged in water and then pressurized to 50 psig. After a period of 10 minutes at this environment the LSC shall be fired at ambient temperature.

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4.5.1.3 Forty-Foot Drop Test. The six assemblies shall be conducted in accordance with test 103 of MIL-STD-331. Two assemblies shall be dropped in the longitudinal axis nose up position, two in the longitudinal axis nose down position and two in the longitudinal axis horizontal position. The assemblies shall be dropped with the end boosters unprotected and no detonations ahall occur.

4.5.1.4 Six-Foot Drop Test. The assemblies with the end boosters protected by shipping covers, shall be dropped from a height of six feet onto a minimum thickness of one foot of reinforced concrete. The assemblies shall not detonate if they are bent the assemblies shall be reformed to the original configuration and shall be capable of satisfactory operation after completion of this test. The drop axis shall be as follows:

(a) One drop on each end with the longitudinal axis perpendicular to the slab.

(b) One drop with longitudinal axis parallel to the slab.

4.5.2 Preproduction Test Report. The contractor shall submit to the procuring activity, for approval, copies of his preproduction test report. The report shall include the test result, all supporting data collected in conducting the tests, and analyses of all failures which occurred. As a supplement the contractor may include, for the procuring activity's consideration, any recommendations for design improvement beyond any corrective action required to pass the test. The data presentation of the test report shall be coordinated with the approval test procedure required by 4.7.

4.5.3 Preproduction Approval. Satisfactory completion of the preproduction test requirements shall be contingent upon the procuring activity's approval of the contractor preproduction test report. Fabrication of production devices prior to such approval is at the contractor's own risk. After completion of the tests, the disposition of the test samples tested shall be as specified in the contract.

4.6 Special Tests. Changes in design and material subsequent to completion of the preproduction tests may require rework and retesting of the devices involved. Such tests as directed by the procuring activity for the purpose of evaluating the changes on the performance and quality of the revised LSC are classified as Special Tests.

4.6.1 Scope of Special Tests. Test procedures previously approved by the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.7 Test Plans and Procedures. Test procedures shall be in accordance with the applicable requirements of MIL-T-18303. The plans and procedures used

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for conducting Preproduction Tests shall be prepared by the contractor and submitted to the procuring activity for approval. The right is reserved by the procuring activity to modify the tests or to require any additional test procedures or details deemed necessary to determine compliance with the requirements of this specification or the contract. The test details, such as the test fixture design, the length of characteristics to be measured, special failure criteria, thermocouple and accelerometer locations, instrumentation description, etc., shall be part of the test procedure to be submitted.

4.8 Rejection and Retest. The assemblies which have been rejected may be reworked, or have parts replaced to correct the defects, and resubmitted for acceptance upon approval of the procuring activity. Before resubmittal, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the procuring activity. After corrections have been made, all tests deemed necessary by the procuring activity shall be repeated. No device that has been fired may be reworked and resubmitted.

NOTE: The following action shall be taken in case of a failure:

(a) Immediately notify the procuring activity's representative.

(b) Determine the cause of failure.

(c) Determine if the failure is a recurring manufacturing problem or design deficiency.

(d) Submit analysis to the procuring activity.

(e) Submit to the procuring activity for approval the proposed corrective action intended to reduce the possibility of the same failure(s) recurring.

5. PREPARATION FOR DELIVERY

5.1 General. Each assembly shall be preserved, packaged, and packed for the level of shipment specified in the contract or purchase order in accordance with MIL-STD-794.

6. NOTES

6.1 Intended Use. The assemblies covered by this specification are intended for use in the Emergency Escape System of Aircraft.

6.2 Precedence of Documents. When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(a) Contract - The contract shall have precedence over any specification.

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(b) This Specification - This specification and its applicable drawing shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(c) Referenced Specifications. Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

6.3 Performance Objectives. Minimum size and weight, simplicity of operation, case of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment.

6.4 Testing Ground Rules.

6.4.1 General Rules. The following ground rules apply to the design verification, preproduction, acceptance and special tests required herein, and supplement those found in the referenced specifications.

(a) A survey shall be conducted on one assembly prior to test to determine the time required to reach thermal stability. This shall be accomplished by monitoring a thermocouple located as closely as possible to the core of the item of greatest thermal inertia.

(b) Throughout the program and after every design change the assembly shall be tested at a unit level and in addition the assembly's and the test fixture(s) shall be tested as an integrated system to assure compatibility.

6.5 Safety Procedures. Standard DOD 4145.26M may be used as a guide.

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